Loh Boon Cheong, Timothy

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Portfolio • GitHub • LinkedIn

Seeking a one-year software engineering internship: May 2027 – May 2028

Education

Singapore Institute of Technology | DigiPen Institute of Technology Singapore
BSc in Computer Science in Real-Time Interactive Simulation — Expected Graduation: 2028
GPA: 3.99/5.00
Relevant Coursework: Data Structures and Algorithms, Operating Systems, Networking, Graphics Programming

Singapore Polytechnic
Diploma in Accountancy — Graduated: 2022

Projects

Hellmaker 2D Engine/Editor — September 2025 – May 2026
Technical Lead | C++, Lua, OpenGL, Dear ImGui, FMOD

- Architected and maintained a modular CMake build system across three targets, using FetchContent with pinned Git tags to manage third-party dependencies and enable consistent builds across developer environments.
- Built the engine's system registrar to manage subsystem lifetimes and coordinate updates across the engine and editor.
- Developed an embedded web server for browser-based frame-time monitoring and runtime inspection.
- Resolved video/audio synchronization drift by using FMOD's audio clock to drive video playback, accounting for output latency and coordinating decoded audio through a mutex-protected sample queue.
- Introduced a PNG-to-BC3/DXT5 import pipeline and direct compressed texture uploads to OpenGL, reducing editor and game loading times from 30 seconds to 0.5 seconds, a 60x speedup.

StrokeNet – Networked Drawing Game — March 2026
Architecture Lead | C++, Winsock, SFML

- Designed data flow between threads in both the client and server, synchronization boundaries around shared state, and binary packet layouts for network commands in a four-person LAN drawing-and-guessing game.
- Designed a UDP protocol combining acknowledged requests and notifications with best-effort drawing updates, using non-blocking Winsock sockets.

x86-64 Assembly Mini Projects — August 2026 - Present
GCC, x86-64 Assembly, Shell, Linux

- Authored multiple square-root approximations and integrated the C standard library's sqrt through an assembly wrapper, applying System V x86-64 calling conventions, register preservation, and stack alignment.
- Created a shell-based benchmark harness using GCC and perf to compare executable runtimes across shared test inputs.

KD-Tree Ray Intersection Implementation — June 2026 – July 2026
C++20 | Spatial Data Structures | Ray Intersection

- Implemented a kdtree within a professor-provided graphics framework, including sweep-based surface-area-heuristic splitting and a compact eight-byte node representation.
- Developed closest-hit ray traversal with AABB interval testing, near-to-far traversal, parallel-ray handling, and early pruning; validated correctness and efficiency against brute-force results using the supplied GoogleTest suite.

Interactive Graphics Demos — January 2026 – May 2026
C++, GLSL, OpenGL, GLFW, GLEW, GLM

- Created nine interactive coursework demos within a supplied graphics framework, covering ray casting, ray tracing, Blinn-Phong lighting, environment mapping, portals, and deferred rendering.
- Built a six-pass cartoon rendering pipeline combining shaded scene output, Gaussian blur, Sobel edge detection, and compositing; implemented normal mapping and G-buffer surface data for deferred shading.

Technical Skills

Languages: C, C++, C#, x86-64 Assembly, Python, Lua, SQL, GLSL
Tools: Linux, GDB, GCC, Git, CMake, Make, GoogleTest, Wireshark, Visual Studio, Jira, Batch Scripting
Libraries and APIs: OpenGL, GLFW, GLEW, GLM, Dear ImGui, Qt (basic), SFML, Winsock, FMOD
Core Areas: Networking, Multithreading, Data Structures and Algorithms, Memory Management, Object-Oriented Programming, Computer Architecture, Shader Programming, Debugging, Unit Testing